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PRIVACY-ENHANCED ONTOLOGIES

A STUDY ON SECURE PROTOCOLS FOR EXPRESSIVE SPARQL
QUERIES OVER ENCRYPTED ONTOLOGIES

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ABSTRACT

1. What is the problem?
2. Why is this problem interesting/challenging?
3. What is the proposed approach/solution?
4. What results (implications/consequences) from the solution?

Keywords: Keyword 1, Keyword 2, Keyword 3, Keyword 4

RESUMO

1. Qual é o problema?
2. Porque é que é um problema interessante/desafiante?
3. Qual é a proposta de abordagem/solução?
4. Quais são as consequências/resultados da solução proposta?

Palavras-chave: Palavra-chave 1, Palavra-chave 2, Palavra-chave 3, ...

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ACRONYMS

DSSE	Dynamic Symmetric Searchable Encryption 3 , 4
FHE	Fully Homomorphic Encryption 3 , 7
GO	Gene Ontology 1
HE	Homomorphic Encryption 2 , 3 , 7
LFHE	Leveled-Fully Homomorphic Encryption 3 , 7
NCI	National Cancer Institute 1
ORAM	Oblivious RAM 3
PEKS	Public Key Encryption with Keyword Searchable 3 , 7
PHE	Partially Homomorphic Encryption 3 , 7
SE	Searchable Encryption 2 , 3 , 7
SHE	Somewhat Homomorphic Encryption 3 , 7
SLA	Service Level Agreement 2
SSE	Symmetric Searchable Encryption 3 , 4 , 7
STE	Structured Symmetric Encryption 3 , 4
W3C	World Wide Web Consortium 6
WWW	World Wide Web 1

INTRODUCTION

1.1 Context

In 2001, the term *Semantic Web* [3] was first used to define what would be the evolution of the [World Wide Web \(WWW\)](#). The traditional *Web* solved the challenge of distributing information at scale and globally, however it was designed for human use. Documents are easily accessible and readable by humans, but for automated agents little more than document retrieval is possible. Contrary to humans, machines cannot reason over the information they access, if they do not know the *semantics of the data*.

The *Semantic Web* addresses this issue, by redesigning the representation of information, in a way that not only preserves meaning and relationships but is also machine-readable. A concept is mapped to a unique identifier, and meaning is encoded in sets of triples, each triple consisting of a subject, predicate and object (similar to an elementary sentence). *Ontologies* provide a shared understanding of a domain by formalizing the specification of relevant concepts, and their relations, in such domain.

This novel method of knowledge representation differs from the previous, centralized way, and greatly simplifies the process of interconnecting different knowledge bases. Centralized representations tend to be more expensive to develop, usually needing to be built from scratch [18]. Furthermore, for sharing, both parties need to agree on the semantics. In contrast, using ontologies, information is semantically linked by default. In the *Semantic Web*, machines can truly communicate, share and reason about distinct knowledge bases.

Knowledge representation is essential in the industry [19] and ontologies help remove ambiguity in terminology. Nowadays, they are being developed and applied in a variety of domains [30] ranging from the health sector¹, to linguistics² and even being used by companies like NASA³ [2].

¹The NCI Thesaurus [25], developed by the [National Cancer Institute \(NCI\)](#) or the [Gene Ontology \(GO\)](#) [12], the world's largest source of information on the functions of genes, are some existing applications.

²The WordNet [15] is a lexical database for English where related words and concepts are semantically linked.

³Observing the visual representation provided by the Linked Open Data Cloud [14] is a good exercise to perceive the dimension of adoption of this new paradigm.

To develop and maintain ontologies, advanced semantic editors are the standard tool for professionals, with Protégé [17] being a very popular example. These provide numerous functionalities, usually based on plugins (e.g. automatic verification of errors and consistency in the design; debugging tools; query answering; graphic visualization of data).

With ontologies gradually growing in dimension, the industry is currently evolving towards the use of collaborative solutions [29, 28] and outsourcing heavy computations, relying on cloud infrastructures or other shared computing platforms. These approaches, albeit inevitable, present new security concerns and engineering challenges.

As of 2021, cloud adoption has risen more than what was previously forecasted, with 50 percent of all enterprise workloads hosted in *public clouds* and predictions of 7 percentage point gain in 2022 [24]. Research on secure cloud computing [6] studies such adoption but also discusses security concerns. Adoption can be explained with the cost efficiency⁴, high scalability, flexibility and availability that cloud provides. Additionally, the pay-per-use model is an easy entry point to the ecosystem.

On the contrary, while delegation of responsibilities is beneficial for costs, it is prejudicial to control. Using cloud services implies trusting its owner. A [Service Level Agreement \(SLA\)](#) defines, in accordance to metrics, the service levels of the system that the vendor should provide (e.g. percentage of availability), while also defining penalties in case of not achieving what was agreed-on. However, the risk of loss of privacy and data breaches still exists. A problem that is especially concerning when sensitive data, such as health records, is being stored in such services. This is much more relevant in *public clouds* where, compared to *private clouds*, trust is distributed between various actors and not on a single organization [6].

In cloud storage services, administrators, or hacker with root rights, might have access to the data [4]. Additionally, these cloud services might outsource computations to third parties whom we do not trust. In this setting, we usually consider the *honest-but-curious*, or *passive*, adversary model. This adversary will follow the protocol properly but keep a record of all its intermediate computations [10]. Furthermore, it cannot do any other action except attempting to learn private information while observing views of the protocol execution [8].

A first approach to secure data in the cloud would be to use *Encryption at Rest*, which naively encrypts all data stored in the untrusted servers. If a user wants to do a write or search operation over the data, they will have to download all the data, and decrypt it. This incurs in high network requirements and is computationally expensive on the client-side, hindering all the benefits of using a cloud service to store the data. Nonetheless, this is the mechanism usually provided by database services [21, 20, 16].

Recently, active research is being done in the fields of [Searchable Encryption \(SE\)](#) and [Homomorphic Encryption \(HE\)](#).

⁴Cloud services reduce implementation and maintenance, power, cooling, floor space, storage and operational costs.

Homomorphic Encryption is a cryptographic method based on the concept of algebra homomorphism, where both decryption and encryption can be seen as an homomorphism of the relations between the plaintext and the ciphertext [1, 9]. **Homomorphic Encryption**, based on the type and number of operations allowed to be done on the encrypted data, can be classified as: **Fully Homomorphic Encryption (FHE)** supporting any number of operations with no restriction on the number of times; **Somewhat Homomorphic Encryption (SHE)** permitting some types of operations, a limited number of times; **Partially Homomorphic Encryption (PHE)** allowing one type of operation with no restriction on the number of times; and **Leveled-Fully Homomorphic Encryption (LFHE)** [5] supporting any number of operations, but with a pre-determined expected depth- L .

Searchable Encryption uses classical encryption schemes to perform encrypted search queries, sequentially over the ciphertext [26], or through the generation of searchable *encrypted indexes* accessible with *tokens*, generated via *trapdoor* functions. **Searchable Encryption** can be divided in **Symmetric Searchable Encryption (SSE)**, when it is based on symmetric key cryptography, and **Public Key Encryption with Keyword Searchable (PEKS)**, when asymmetric key algorithms are used. These techniques may leak some information about the *indexes*, *search pattern*, and *access pattern* during execution. Most approaches leak, at least, the *search* and *access patterns* [4].

Besides **HE** and **SE**, **Oblivious RAM (ORAM)** [11] could theoretically be used as an alternative. However, **ORAM** and **HE** are not efficient enough, when used exclusively, to be feasible yet [4] in real world applications. Also, **SSE** tends to be more practical than **PEKS**, due to the efficiency comparison between asymmetric and symmetric cryptographic schemes [22].

1.2 Problem

Ontologies usually only contain public, and non-sensitive data, but sometimes they are used alongside electronic health records and clinical decision systems. Furthermore, they can hold private value, due to the investment of time and money in their development.

Taking into considerations the efficiency and security trade-offs from the various alternative techniques, the **SSE** approach seems to be the most practical solution for protecting ontologies in untrusted servers. However, most **SSE** protocols only consider *unstructured* collections of documents with no semantic relationships. More recently, Chase and Kamara, on their work on **Structured Symmetric Encryption (STE)**, generalized index-based **SSE** to support queries on *structured* data [7].

Ontologies can be represented as graphs and **STE** supports various types of query operations in graph-structured data. Nevertheless, **STE** with direct support for the query expressiveness required for searches over ontologies has not yet been studied, more specifically for languages like *SPARQL* [23, 27].

Furthermore, write operations that are supported in **Dynamic Symmetric Searchable Encryption (DSSE)** [13], have not yet been studied in the context of ontologies.

Additionally, information leakage can be a problem while trying to guaranty privacy of the topology of an ontology.

To conclude, we will focus on answering the following questions:

1. *How to support expressive queries and other operations in privacy-enhanced encrypted ontologies, using techniques such as [Structured Symmetric Encryption](#) and [Dynamic Symmetric Searchable Encryption](#)?*
2. *Besides preserving data privacy, how can we minimize information leakage and avoid possible exposure of the topology of the encrypted ontologies?*

1.3 Objectives

The main objective of this thesis will be to provide answers to the two questions defined in section 1.2 through a problem driven approach and guide future work in this field of research. We will first study a static [STE](#) approach focusing on support for *SPARQL* queries in a multi-user setting, later expanding to a dynamic setting. Finally, we will study how to minimize information leakage about the topology of the encrypted data.

1.4 Contributions

Our contributions will be the proposal of a protocol that extends upon [STE](#) and supports expressive and secure *SPARQL* queries over privacy-enhanced encrypted ontologies. We now present a summary of the contributions:

SSE M/M Protocol for Privacy-Enhanced Ontologies A novel multi-user protocol, based on [STE](#), that supports *SPARQL* queries over encrypted ontologies, stored in untrusted servers.

P1 - Static Setting A protocol implementation for a static setting, supporting only search operations, based on [STE](#).

P2 - Dynamic Setting A protocol implementation for a dynamic setting, supporting search, write and delete operations, based on [STE](#) and [SSE](#).

1.5 Document Organization

The remaining of the document will be organized as follows:

Chapter 2 Analyses related work on *Semantic Web*, *Ontologies* and *Searchable Encryption*.

Chapter 3 Presents requirements and informally defines the protocols and system models. Furthermore, in this chapter we describe our implementation and testing approaches.

Chapter 4 Describes the work plan of the elaboration phase tasks.

RELATED WORK

2.1 Semantic Web: A layered architecture

In section 1.1, we have contextualized the main philosophy and design principles of the *Semantic Web* for the most granular level (a triple of subject, predicate and object). Additionally, the *Semantic Web* can also be seen as a stack of various layers, figure 2.1. Most of these layers are standardized due to the collective work of the [World Wide Web Consortium \(W3C\)](#).

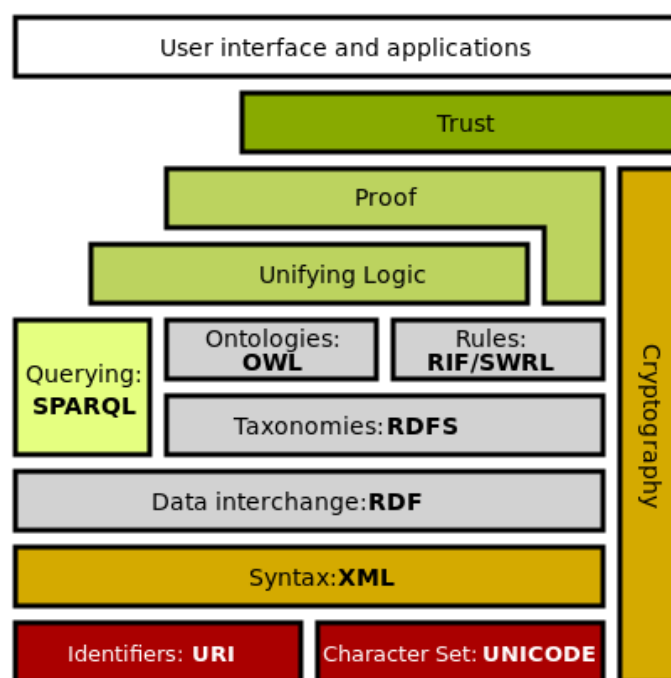


Figure 2.1: The *Semantic Web* stack [31]

2.1.1 RDF: Resource Description Framework

2.1.2 RDFS: RDF Schema

2.1.3 OWL2: Web Ontology Language

2.1.4 SPARQL

2.1.5 Tools & Development Software for Ontologies

2.2 Secure Computations

2.2.1 Homomorphic Encryption

Homomorphic Encryption is a cryptographic method based on the concept of algebra homomorphism, where both decryption and encryption are seen as homomorphisms of the relations between the plaintext and the ciphertext [1, 9]. **HE** allows for computations done over encrypted data to be preserved after decryption of the ciphertext. An encryption scheme is homomorphic if such operations preserve additive or multiplicative properties. **HE** is classified in four main types: **FHE** supporting any number of operations with no restriction on the number of times; **SHE** permitting some types of operations, a limited number of times; **PHE** allowing one type of operation with no restriction on the number of times; and **LFHE** [5] supporting any number of operations, but with a pre-determined expected depth- L .

2.2.2 Searchable Encryption

Searchable Encryption uses classical encryption schemes to perform encrypted search queries. **Searchable Encryption** can be divided in **SSE**, when it is based on symmetric key cryptography, and **PEKS**, when public key algorithms are used. These techniques may leak some information about the *indexes*, *search pattern*, and *access pattern* during execution.

2.2.3 Symmetric Searchable Encryption

2.2.3.1 Structured Searchable Encryption

2.2.3.2 Dynamic Symmetric Searchable Encryption

2.3 Summary

APPROACH TO ELABORATION

3.1 System Requirements

3.2 Privacy Guaranties

3.3 Architecture & System Model

3.3.1 Implementation Guidelines

3.3.2 Validation

WORK PLAN

BIBLIOGRAPHY

- [1] A. Acar et al. “A survey on homomorphic encryption schemes: Theory and implementation”. In: vol. 51. 2018. DOI: [10.1145/3214303](https://doi.org/10.1145/3214303) (cit. on pp. 3, 7).
- [2] N. Ashish. “Semantic-Web Technology: Applications at NASA”. In: *Dagstuhl Seminar Proceedings*. 2005 (cit. on p. 1).
- [3] T. Berners-Lee, J. Hendler, and O. Lassila. *The semantic web*. Vol. 284. 2001. DOI: [10.1038/scientificamerican0501-34](https://doi.org/10.1038/scientificamerican0501-34) (cit. on p. 1).
- [4] C. Bösch et al. “A Survey of Provably Secure Searchable Encryption”. In: *ACM Comput. Surv.* 47.2 (Aug. 2014), pp. 1–51. ISSN: 0360-0300. DOI: [10.1145/2636328](https://doi.org/10.1145/2636328) (cit. on pp. 2, 3).
- [5] Z. Brakerski, C. Gentry, and V. Vaikuntanathan. “(Leveled) Fully Homomorphic Encryption without Bootstrapping”. In: *ACM Trans. Comput. Theory* 6.3 (July 2014), pp. 1–36. ISSN: 1942-3454. DOI: [10.1145/2633600](https://doi.org/10.1145/2633600) (cit. on pp. 3, 7).
- [6] M. Carroll, A. Van Der Merwe, and P. Kotzé. “Secure cloud computing: Benefits, risks and controls”. In: *2011 Information Security for South Africa - Proceedings of the ISSA 2011 Conference*. 2011. DOI: [10.1109/ISSA.2011.6027519](https://doi.org/10.1109/ISSA.2011.6027519) (cit. on p. 2).
- [7] M. Chase and S. Kamara. “Structured encryption and controlled disclosure”. In: *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. Vol. 6477 LNCS. 2010. DOI: [10.1007/978-3-642-17373-8_33](https://doi.org/10.1007/978-3-642-17373-8_33) (cit. on p. 3).
- [8] D. Evans, V. Kolesnikov, and M. Rosulek. “A Pragmatic Introduction to Secure Multi-Party Computation”. In: *Foundations and Trends® in Privacy and Security* 2.2-3 (2018), pp. 70–246. ISSN: 2474-1558. DOI: [10.1561/33000000019](https://doi.org/10.1561/33000000019) (cit. on p. 2).
- [9] C. Fontaine and F. Galand. “A survey of homomorphic encryption for nonspecialists”. In: *Eurasip Journal on Information Security* 2007 (2007). ISSN: 2510-523X. DOI: [10.1155/2007/13801](https://doi.org/10.1155/2007/13801) (cit. on pp. 3, 7).

-
- [10] O. Goldreich. *Foundations of Cryptography: Volume 2, Basic Applications*. en. Cambridge University Press, 2001. ISBN: 9780521830843 (cit. on p. 2).
 - [11] O. Goldreich and R. Ostrovsky. “Software protection and simulation on oblivious RAMs”. In: *J. ACM* 43.3 (May 1996), pp. 431–473. ISSN: 0004-5411. DOI: [10.1145/233551.233553](https://doi.org/10.1145/233551.233553) (cit. on p. 3).
 - [12] M. A. Harris et al. “The Gene Ontology (GO) database and informatics resource”. en. In: *Nucleic Acids Res.* 32.Database issue (Jan. 2004), pp. D258–61. ISSN: 0305-1048, 1362-4962. DOI: [10.1093/nar/gkh036](https://doi.org/10.1093/nar/gkh036) (cit. on p. 1).
 - [13] S. Kamara, C. Papamanthou, and T. Roeder. “Dynamic searchable symmetric encryption”. In: *Proceedings of the 2012 ACM conference on Computer and communications security*. CCS ’12. Raleigh, North Carolina, USA: Association for Computing Machinery, Oct. 2012, pp. 965–976. ISBN: 9781450316514. DOI: [10.1145/2382196.2382298](https://doi.org/10.1145/2382196.2382298) (cit. on p. 3).
 - [14] J. P. McCrae. *The linked open data cloud*. en. <https://lod-cloud.net/>. Accessed: 2022-2-8 (cit. on p. 1).
 - [15] G. A. Miller. “WordNet: a lexical database for English”. In: *Commun. ACM* 38.11 (Nov. 1995), pp. 39–41. ISSN: 0001-0782. DOI: [10.1145/219717.219748](https://doi.org/10.1145/219717.219748) (cit. on p. 1).
 - [16] MongoDB, Inc. *MongoDB Manual: Encryption at rest*. en. <https://docs.mongodb.com/manual/core/security-encryption-at-rest/>. Accessed: 2022-2-9 (cit. on p. 2).
 - [17] M. A. Musen and Protégé Team. “The Protégé Project: A Look Back and a Look Forward”. en. In: *AI Matters* 1.4 (June 2015), pp. 4–12. ISSN: 2372-3483. DOI: [10.1145/2757001.2757003](https://doi.org/10.1145/2757001.2757003) (cit. on p. 2).
 - [18] R. Neches et al. “Enabling technology for knowledge sharing”. In: *AI Magazine* 12.3 (1991). ISSN: 0738-4602 (cit. on p. 1).
 - [19] N. Noy et al. “Industry-scale knowledge graphs: Lessons and challenges”. In: *Commun. ACM* 62.8 (2019). ISSN: 0001-0782, 1557-7317. DOI: [10.1145/3331166](https://doi.org/10.1145/3331166) (cit. on p. 1).
 - [20] Oracle Corporation and/or its affiliates. *MySQL :: MySQL 8.0 reference manual :: 15.13 InnoDB data-at-rest encryption*. en. <https://dev.mysql.com/doc/refman/8.0/en/innodb-data-encryption.html>. Accessed: 2022-2-9 (cit. on p. 2).
 - [21] Oracle Corporation and/or its affiliates. *MySQL enterprise encryption*. en. <https://www.mysql.com/products/enterprise/encryption.html>. Accessed: 2022-2-9 (cit. on p. 2).
 - [22] W. Pearson. *Cryptography and Network Security: Principles and Practice, Global Edition*. 7th ed. Pearson Education, 2017. ISBN: 9781292158587 (cit. on p. 3).

- [23] E. Prud'hommeaux and A. Seaborn. *SPARQL Query Language for RDF*. <https://www.w3.org/TR/rdf-sparql-query/>. Jan. 2008 (cit. on p. 3).
- [24] Security Compass. *The 2021 State of Cloud Adoption*. en. <https://resources.securitycompass.com/reports/2021-state-of-cloud-adoption>. Accessed: 2022-2-9. Sept. 2021 (cit. on p. 2).
- [25] N. Sioutos et al. "NCI Thesaurus: a semantic model integrating cancer-related clinical and molecular information". en. In: *J. Biomed. Inform.* 40.1 (Feb. 2007), pp. 30–43. ISSN: 1532-0464, 1532-0480. DOI: [10.1016/j.jbi.2006.02.013](https://doi.org/10.1016/j.jbi.2006.02.013) (cit. on p. 1).
- [26] D. X. Song, D. Wagner, and A. Perrig. "Practical techniques for searches on encrypted data". In: *Proceeding 2000 IEEE Symposium on Security and Privacy*. S P 2000. May 2000, pp. 44–55. DOI: [10.1109/SECPRI.2000.848445](https://doi.org/10.1109/SECPRI.2000.848445) (cit. on p. 3).
- [27] The W3C SPARQL Working Group. *SPARQL 1.1 Overview*. <https://www.w3.org/TR/sparql11-overview/>. Mar. 2013 (cit. on p. 3).
- [28] T. Tudorache, J. Vendetti, and N. F. Noy. "Web-Protégé: A lightweight OWL ontology editor for the web". In: *CEUR Workshop Proceedings*. Vol. 432. 2009 (cit. on p. 2).
- [29] T. Tudorache et al. "Supporting collaborative ontology development in Protégé". In: *Lect. Notes Comput. Sci.* 5318 LNCS (2008), pp. 17–32. ISSN: 0302-9743, 1611-3349. DOI: [10.1007/978-3-540-88564-1_2](https://doi.org/10.1007/978-3-540-88564-1_2) (cit. on p. 2).
- [30] W3C. *Semantic Web Case Studies and Use Cases*. en. <https://www.w3.org/2001/sw/sweo/public/UseCases/>. Accessed: 2022-2-8 (cit. on p. 1).
- [31] Wikipedia contributors. *Semantic web stack*. https://en.wikipedia.org/wiki/File:Semantic_web_stack.svg. Accessed: NA-NA-NA (cit. on p. 6).